Cover Letter

November 1, 2026

Faculty Search Committee
Department of Physics & Engineering
The University of Scranton
Scranton, Pennsylvania

Re: Assistant Professor of Electrical/Computer Engineering, Posting F00625P

Dear Members of the Search Committee:

I am applying for the tenure-track Assistant Professor position in Electrical/Computer Engineering beginning in August 2027. I earned my PhD in Electrical and Computer Engineering from the University of Oklahoma, where I investigated machine learning for wireless-network modeling, optimization, and resilience. I previously taught undergraduates in a technology department and have worked in engineering research at AT&T Labs, Futurewei, and AT&T. The opportunity to combine undergraduate teaching, hands-on engineering, and a research program with students at the University of Scranton is especially appealing to me.

At the University of Lahore, I taught Computer Programming, Communication Systems, and Wireless Communication in the Department of Technology. I learned that students gain confidence when they can move from an equation or a piece of code to an experiment they can explain. At the University of Engineering and Technology, Taxila, I supported communications laboratories, coursework, and assessment; at COMSATS University Islamabad, I mentored students working on research projects and theses. My teaching also extends to tutoring students outside the university classroom. These experiences made me attentive to different levels of preparation and to the value of precise, timely feedback.

My strongest preparation for Scranton's EE and CE programs is in programming, communication and wireless systems, signal analysis, engineering computing, and applications of AI and machine learning. I am particularly interested in courses and laboratories where students can test a theoretical prediction using MATLAB, Python, or measured signals. Scranton's computer engineering program emphasizes the integration of software, hardware, design, and laboratory experience, culminating in a substantial robotics capstone. I would welcome opportunities to help students develop the programming, sensing, validation, and data-analysis aspects of such work while preparing carefully for other courses assigned within the department. I do not assume that experience in wireless AI replaces specialized preparation in power electronics or digital hardware.

My scholarship addresses a practical question: how can an intelligent engineering system remain dependable when the data are limited and operating conditions change? My doctoral projects investigated data-efficient propagation prediction, generative and transfer learning, and detecting or compensating for cellular network outages. Related work with smartwatch sensor signals extends the same questions to user-dependent activity recognition. I envision an active program in reliable AI for wireless and sensor-based systems, organized around reproducible simulation, open or appropriately authorized measurements, and well-scoped undergraduate projects. A student could begin with a baseline model, test a controlled change, and ultimately take ownership of a result suitable for a capstone, poster, or scholarly paper.

I am also drawn to Scranton's Jesuit emphasis on education that develops technical ability alongside judgment, responsibility, and service to others. In engineering, that means helping students ask who might be affected by a design failure, how evidence supports a decision, and what limitations must be communicated. I would bring that perspective to classroom teaching, mentoring, and collaborative work on course and ABET assessment.

Thank you for considering my application. I would welcome the chance to contribute to Scranton's undergraduate engineering programs and to develop research opportunities in which students participate in meaningful engineering inquiry.

Sincerely,

Waseem Raza, PhD

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